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Course Code: ITA0305

Course Name: Mobile Computing

Experiment 1: Free-Space Path Loss and Received Power

Program Code:

```
import numpy as np
import matplotlib.pyplot as plt
d = np.linspace(0.1, 5, 100)  # distance in km
f = 900                  # frequency in MHz
pt = 30                  # transmit power in dBm
# Calculate FSPL
fspl = 20*np.log10(d) + 20*np.log10(f) + 32.44
# Calculate received power
pr = pt - fspl
# Display output
print("----- Free Space Path Loss and Received Power -----")
print("Frequency:", f, "MHz")
print("Transmit Power:", pt, "dBm")
print("\nAt Distance 0.1 km:")
print("FSPL:", round(fspl[0], 2), "dB")
print("Received Power:", round(pr[0], 2), "dBm")
print("\nAt Distance 5 km:")
print("FSPL:", round(fspl[-1], 2), "dB")
print("Received Power:", round(pr[-1], 2), "dBm")
```

```
print("\nResult: FSPL and Received Power calculated successfully.")

# Plot graph

plt.plot(d, pr, color='blue', label='Received Power')

plt.xlabel("Distance (km)")

plt.ylabel("Received Power (dBm)")

plt.title("Free-Space Path Loss at 900 MHz")

plt.legend()

plt.grid(True)

plt.show()
```

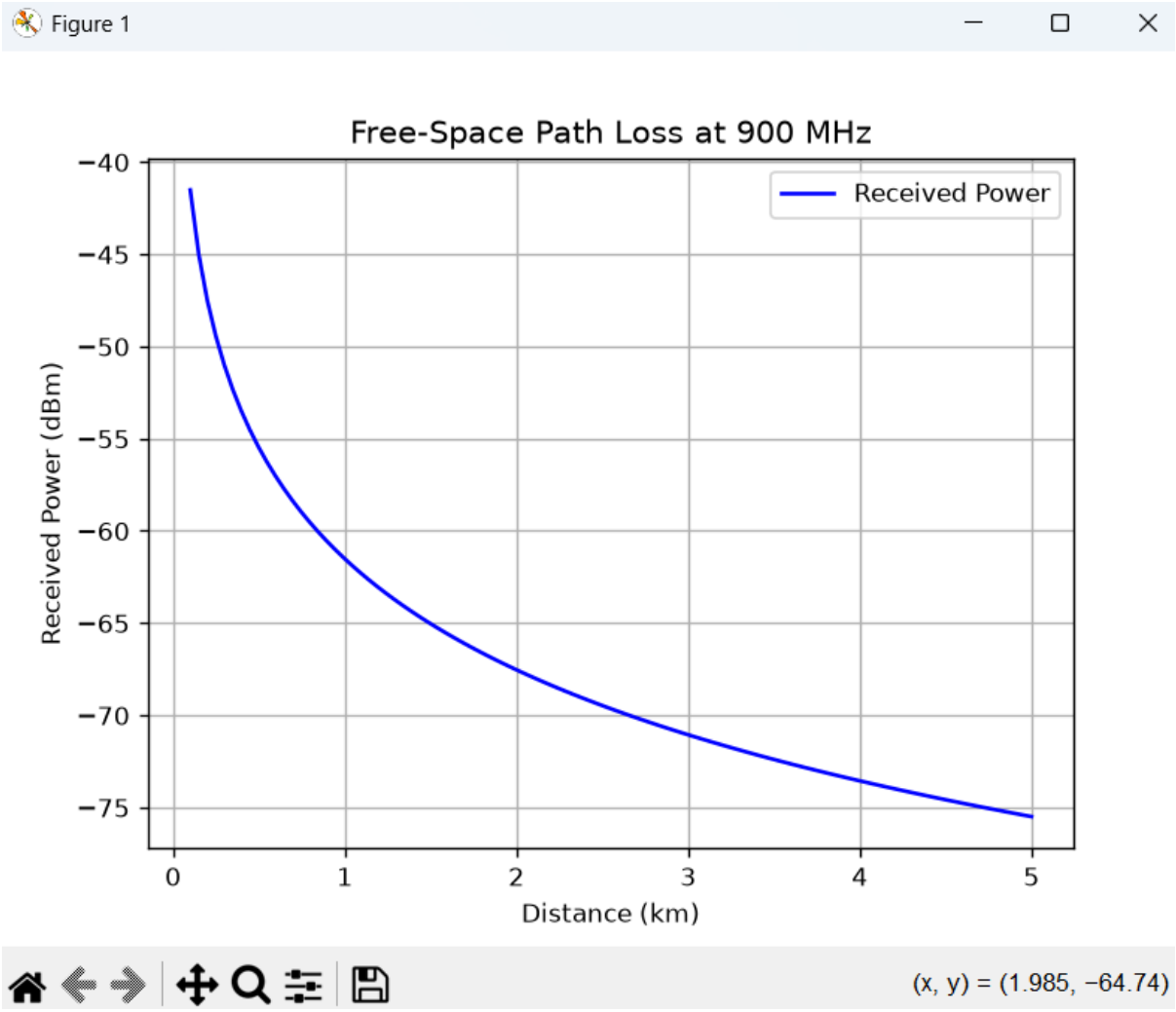
```
======== RESTART: D:/knowledge_Engineering/mobile computing/program1.p

======== RESTART: D:/knowledge_Engineering/mobile computing/program1.p
----- Free Space Path Loss and Received Power -----
Frequency: 900 MHz
Transmit Power: 30 dBm

At Distance 0.1 km:
FSPL: 71.52 dB
Received Power: -41.52 dBm

At Distance 5 km:
FSPL: 105.5 dB
Received Power: -75.5 dBm

Result: FSPL and Received Power calculated successfully.
```


Figure 1
Free-Space Path Loss at 900 MHz
Received Power
Received Power (dBm)
−40
−45
−50
−55
−60
−65
−70
−75
0
1
2
3
4
5
Distance (km)
(x, y) = (1.985, −64.74)